Module-2

① Explain feature selection compare filter and wrapper methods.

Feature selection :- It is the process of selecting the most relevant and useful features is called feature selection.

In machine learning a data set may contain a large no. of features not all features are useful for predicting the target variable.

Ex:- Suppose we want to predict whether a student will pass an exam. The dataset may contain :-

- Attendance
- Internal marks
- Study hours
- Previous marks
- Student name
- Roll number
- Favourite colour

Objectives of feature selection :-

1. Reduce the no. of features in the dataset.
2. Remove irrelevant & medventant features.
3. Improve model performance
4. Reduce overfitting.

Types of feature selection methods :-

1. filter Methods
2. wrapper methods
3. Embedded methods

- Required less data preprocessing.
- Handles numerical & categorical data.

Disadvantages :-

- They can easily overfit training data.
- very deep trees can become complex.
- Large trees may be difficult to implement

(3) Explain principal component analysis (PCA) & its applications ?

PCA :-

It is a dimensionally reduction technigue used in machine learning and statistics to reduce the no. of features in a dataset while presentation as much important as possible. In a dataset many features may contain similar or redundant information, PCA transforms the original features into a new set of variables.

Why PCA Needed :-

→ Increase computation cost.
→ Make visualization difficult.
→ Increase the possibility of overfitting.
→ Make the dataset more complex.

Advantages :-

1. Reduce dimensionality
2. Reduce computational complexity
3. Removes Redundancy
4. Helps redundancy
5. Can reduce noise

Applications :-

1. Data visualization
2. Image processing
3. Machine Learning
4. Face recognization.

Limitations :-

1. It is sensitive of feature scaling.
2. PCA can make the original features harder to interrupt.
3. PCA is mainly designed to capture linear relationship.